



PREPARED FOR NRG | EXECUTIVE WORKING DISCUSSION

PROTECTING TIME TO POWER

A working model for integrated electrical delivery across NRG's generation buildout.

**PRESERVE UPSTREAM CERTAINTY
THROUGH THE ELECTRICAL DELIVERY
CHAIN.**

Stephan Hardt | Director, Power Generation Solutions | Southwire



WHY NOW

NRG's growth agenda creates three distinct electrical-delivery environments.

1.5 GW

TEF PORTFOLIO

One project operating; two targeting mid-2028.

5.4 GW

NEWBUILD DEVELOPMENT

Turbine and EPC access supports a runway through 2032.

25.8 GW

OPERATING FLEET

Approximately 1-2 GW of upgrades under evaluation.

DIFFERENT ASSET STATES. ONE REQUIREMENT.

Preserve schedule certainty through energization.

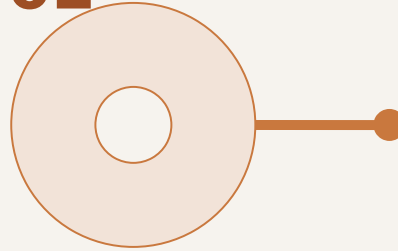
Specifications, capacity, releases, reels, package interfaces and field readiness still have to converge on COD.

THREE RISKS

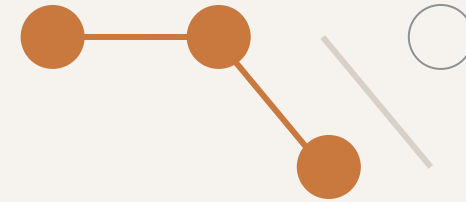
Three execution risks can erode certainty between design and energization.

01**LABOR PRESSURE**

Long pulls, dense terminations and compressed turnover windows concentrate labor when flexibility is lowest.

02**MATERIAL + INSTALLATION FIT**

Ratings, routes, reel lengths, accessories and releases must match the workspace—not only the BOM.

03**FRAGMENTED COORDINATION**

Civil, electrical, OEM and contractor decisions can converge after routing and procurement choices are locked.

LATE HANDOFFS BECOME SCHEDULE AND COMMISSIONING EXPOSURE.

TIME TO POWER IS AN ELECTRICAL EXECUTION PROBLEM BEFORE IT IS A CABLE-BUYING PROBLEM.

- Interfaces cross packages, zones, contractors and turnover paths.
- Late material decisions return as handling, rework and commissioning exposure.
- Managing the workstream as a system can protect schedule better than isolated purchase-order optimization.



PLANT-ZONE COVERAGE

Six clusters map a routed 3×1 electrical reference model.

1, 16

GRID + SITE BACKBONE

Switchyard, GSU/grid tie and MV corridors
 HV/EHV · MV · protection/control · fiber ·
 grounding

2, 3, 10

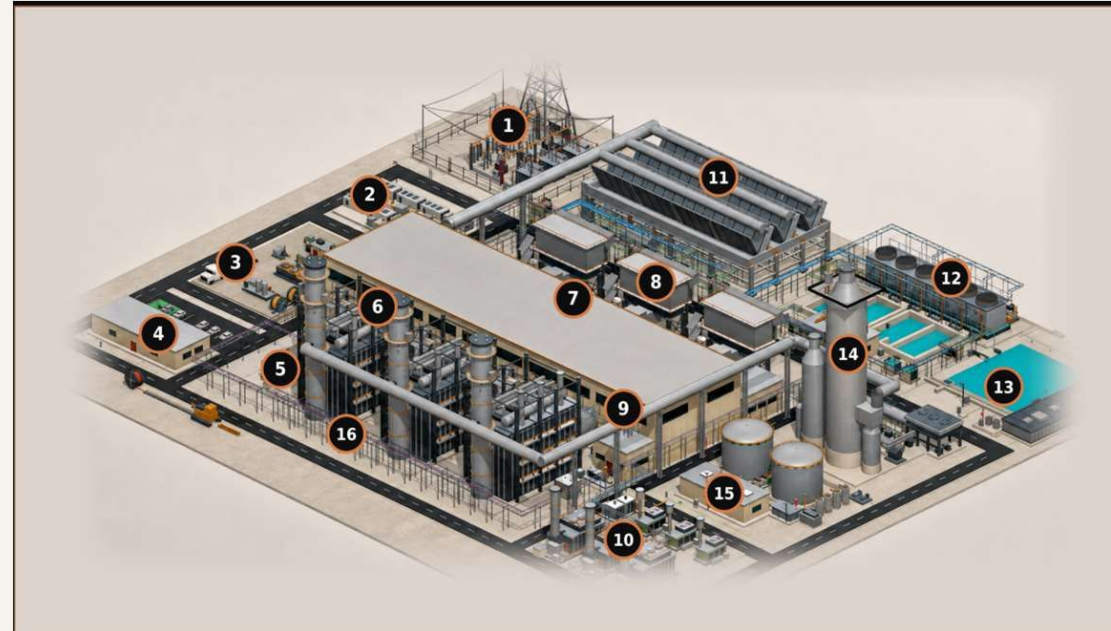
MODULAR + LOGISTICS

BESS, reel staging, prefab/skids and modular
 power
 DC/MV/LV · controls · fiber · engineered sets

4, 5, 9

BUILDINGS + DISTRIBUTION

Control building, e-house and MCC/VFD/UPS
 MV/LV/VFD · essential AC/DC · life safety ·
 BAS/data



6, 7, 8

GENERATION ISLAND

HRSG, turbine hall and GT inlet
 MV/LV/VFD · I&C · F&G/ESD · CEMS · heat
 trace

11, 12, 13

HEAT REJECTION + WATER

ACC, cooling and water/wastewater
 Repeated fan/pump power · VFD · controls ·
 wet runs

14, 15

OPTIONAL + AUXILIARY

Carbon capture and gas metering
 Classified power/control · instrumentation ·
 F&G · fiber

CABLE CLASSES

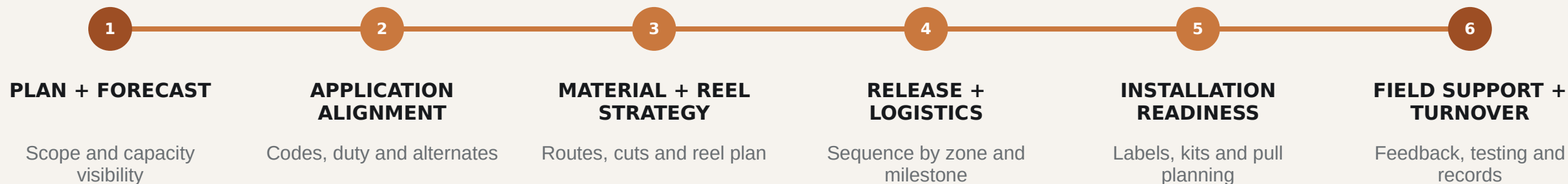
HV/EHV | MV 5-35 kV | LV 600 V | VFD | essential AC/DC | I&C | fiber/network | life safety | grounding

Illustrative—not NRG/IFC. Excludes generator-output bus, OEM wiring and contractor methods.

ONE COORDINATED FLOW

A coordinated electrical delivery system aligns decisions from planning through turnover.

Southwire connects application support, material planning, sequenced logistics and field readiness—without changing project accountability.



SOUTHWIRE COORDINATION RAIL Shared data | governed handoffs | field feedback

NRG governance, EPC/EOR/OEM design authority and contractor means and methods remain intact.

FIELD-ENGINEER WHAT IS UNIQUE. PREFABRICATE WHAT REPEATS.

THE OUTCOME

The value to NRG is fewer execution surprises—not more supplier services.

Three owner lenses matter: schedule exposure, installed effort and lifecycle continuity.

**ALIGN APPLICATIONS +
INTERFACES EARLY**



Fewer late changes and handoff failures

GATE CAPACITY + RELEASES



More predictable material availability and sequencing

**PLAN REELS + REPEATABLE SETS
TO THE WORKFACE**



Less handling and avoidable rework

**CARRY FIELD RECORDS THROUGH
TURNOVER**



Cleaner commissioning and lifecycle continuity

OUTCOME HYPOTHESES TO VALIDATE ON ONE LIVE NRG SCOPE.

OWNER ECONOMICS

Cable is a small cost category with outsized execution consequences.

The owner case is not unit price alone; it is how electrical decisions influence COD readiness, installed effort, capital certainty and lifecycle continuity.

01 COD / REVENUE START

Capacity visibility, release timing and installation readiness can influence energization milestones.

02 INSTALLED COST

Routes, reels, cuts, pulls, congestion and rework often matter more than purchase price alone.

03 CAPITAL CERTAINTY

Earlier scope visibility supports commodity planning, material allocation and contingency discipline.

04 AVAILABILITY / LIFECYCLE

Turnover records, spares, condition screens and replacement planning support long-term continuity.

ILLUSTRATIVE ADJACENT-MARKET BENCHMARK | NOT AN NRG RESULT

Adjacent experience indicates where to test—not what NRG will save.

Field hours per electrical spine in one repetitive modular-generation scope

4,400

OBSERVED FIELD HOURS
SITE-BUILT BASELINE



~880

MODELED FIELD HOURS
WITH FACTORY-BUILT ASSEMBLY

WHAT IT SUGGESTS

Stable, repeatable interfaces may be candidates for offsite work.

WHAT IT DOES NOT PROVE

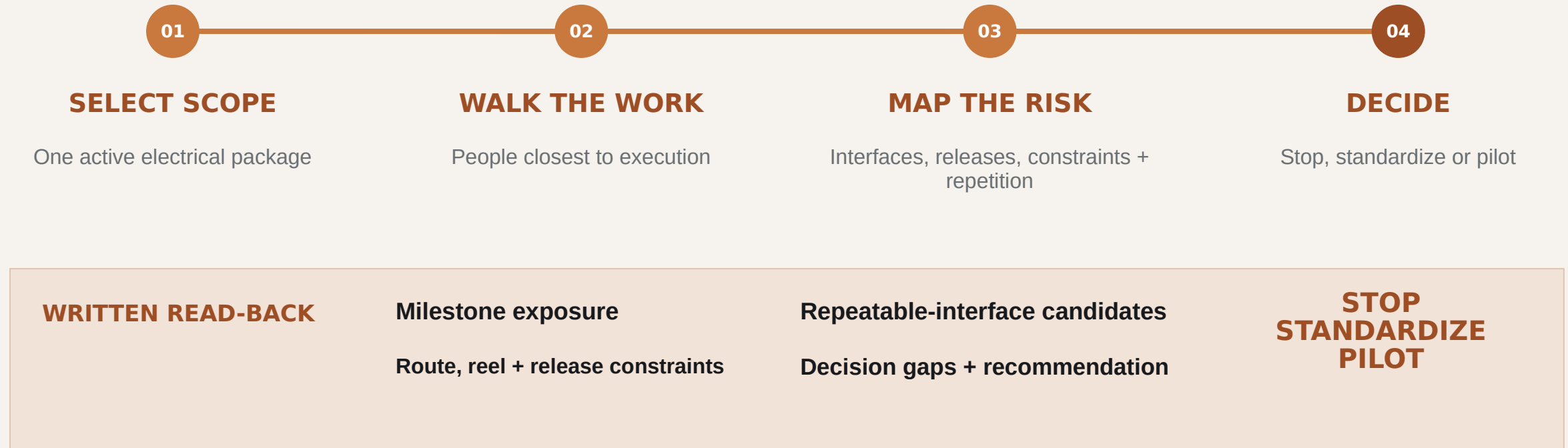
Net NRG savings, schedule reduction or critical-path removal.

Rebuild the case with NRG scope, labor rates, schedule logic and acceptance criteria.

ONE LIVE SCOPE

One live scope can prove—or disprove—the case before NRG expands it.

NRG selects the starting scope: one live package, conversion/uprate screen or planned-outage replacement.



No presumption that the answer is prefab—or that Southwire belongs in the solution.



THE ASK

ONE SITE WALK. ONE HOUR. NO SALES PRESENTATION.

Choose one live electrical scope and bring the people closest to execution.

INPUT

One live scope

OUTPUT

Written execution-risk read-back

DECISION

Stop, standardize or define a bounded pilot

NO PRICING EXERCISE. NO BROAD COMMERCIAL COMMITMENT.

SOURCE DISCIPLINE

NRG context is current, attributed and non-additive.

1.5 GW TEF PORTFOLIO

One operating project and two projects targeting mid-2028. Do not describe the entire portfolio as under construction.

5.4 GW NEWBUILD DEVELOPMENT

Turbine and EPC access supports a runway through 2032. Do not call all 5.4 GW committed or under construction.

25.8 GW OPERATING FLEET

NRG's reported operating-generation fleet after the LS Power acquisition.

~1-2 GW UPGRADE POTENTIAL

CT-to-CCGT conversions and traditional uprates are under evaluation—not a committed pipeline.

DO NOT SUM THESE FIGURES: THEY REPRESENT DIFFERENT ASSET STATES AND OVERLAP CONCEPTUALLY.

DECISION RIGHTS

Clear decision rights preserve existing project accountability.

NRG	Owner priorities, portfolio standards, risk tolerances, acceptance criteria and investment decisions.
EPC / EOR	Design authority, calculations, specifications, routing basis and approved technical design.
OEM	Equipment-package interfaces, internal wiring, approved terminations and warranty boundaries.
ELECTRICAL CONTRACTOR	Installation means and methods, pull planning, workface execution, testing support and field feedback.
SOUTHWIRE	Application support, material and capacity visibility, engineered reel strategy, sequenced delivery and selected field support.

Southwire connects selected decisions; it does not replace owner, EPC, OEM or contractor accountability.

OBSERVED, MODELED, VALIDATED

Evidence is credible only when its boundaries stay visible.

OBSERVED + PUBLISHED

- Name the scope, source, permission and unit of measure.
- Separate measured performance from arithmetic and inference.
- Keep customer- and line-specific conditions attached to each result.

MODELED + DIRECTIONAL

- State every input, exclusion and scope-equivalence assumption.
- Include engineering, factory work, freight, FAT and residual field work.
- Test whether the activity leaves the critical path before treating elapsed-time compression as schedule impact.

DO NOT TURN A SCREENING MODEL INTO AN NRG OUTCOME.

Release gate: source permission, scope equivalence and NRG-specific validation approved.